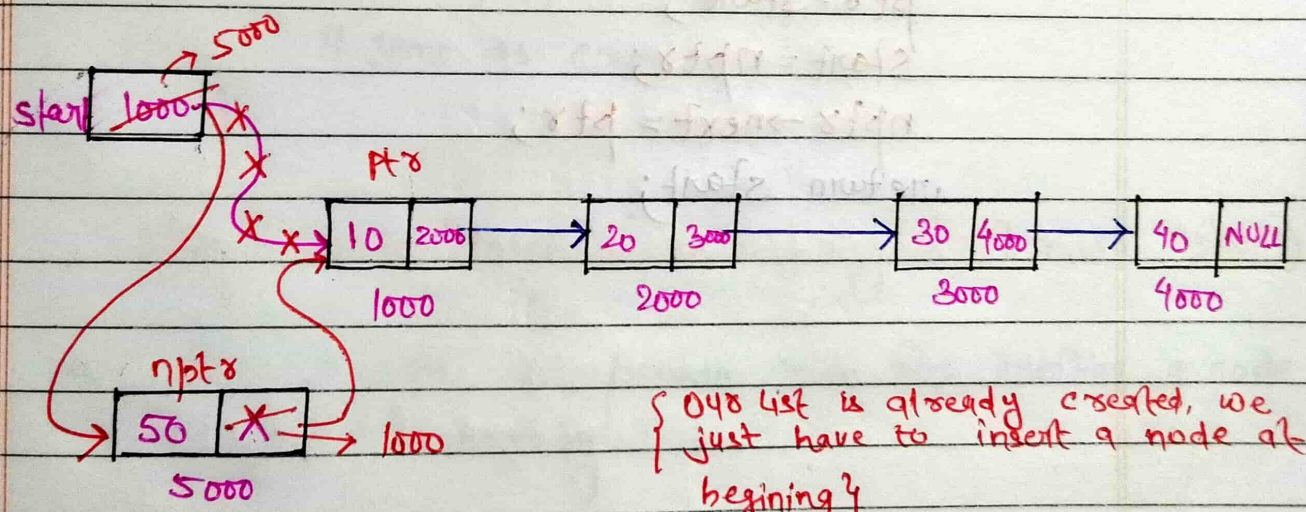
→ Insertion into a linked list

Insertion in a linked list means adding a new node at a specific position while maintaing the structure of the list.

There are three main types of insertion in a singly linked list :

1.) Insertion at the begining :

The new node become the first node.



{ old list is already created, we just have to insert a node at begining }

Algorithm :-

Step 1: Creat nptr [new_node]
2. Set ptr = start
3. Set start = nptr
4. Set nptr → next = ptr
5. Exit

Akash

→ pseudo code for inserting a node at begining;

```
struct node *ptr, *nptr;
int val;
// creating node to insert
nptr =(struct node *)malloc(sizeof(struct node));
printf("enter value to insert");
scanf("%d", &val);
nptr->data = val;
nptr->next = NULL;
ptr=start;
start = nptr;
nptr->next = ptr;
return start;
```

Akash

// c program for inserting a node at begining

```
#include<stdio.h>
#include<stdlib.h>
struct node
{
    int data;
    struct node *next;
};

struct node *create_list(struct node *start)
{
    // same as creating_list
}

struct node *insertion_at_begining(struct node *start)
{
    // same as pseudo code for inserting a node
    // at begining
}

struct node *display_list(struct node *start)
{
    // same as display list
}

int main()
{
    struct node *start = NULL;
```

Akash

```
    start = create_list (start);
    start = insertion_at_begining(start);
    start = display_list(start);
    return 0;
}
```

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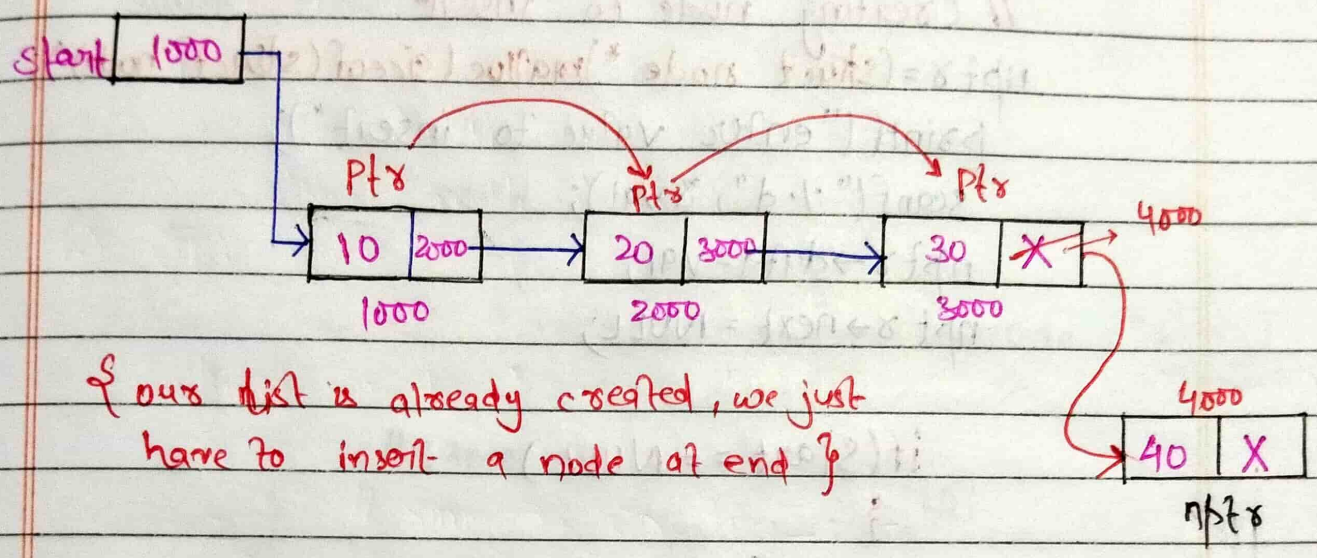
(ii) Insert at the end :

The new node is added after the last node.



{our list is already created, we just have to insert a node at end}

Algorithm :

Step 1. Create nptr [new_node]

2. IF (Start == NULL)

2.1 start = nptr

3. Else

3.1 set ptr = start

3.2 Repeat until ptr → next != NULL

3.2.1 set ptr = ptr → next

3.3. ptr → next = nptr

4. Exit

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pseudo code for insert a new node at last

```
struct node *ptr, *nptr;
int val;
// creating node to insert
nptr=(struct node *)malloc(sizeof(struct node));
printf("enter value to insert");
scanf("%d", &val);
nptr->data=val;
nptr->next = NULL;

if(start == NULL)
{
    start = nptr;
}

else
{
    ptr = start;
    while(ptr->next != NULL)
    {
        ptr = ptr->next;
    }
    ptr->next = nptr;
}
return start;
```

Akash

```
// C program for inserting a node at last

#include<stdio.h>
#include<stdlib.h>
struct node
  {
      int data;
      struct node *next;
  };
struct node *create_list(struct node *start)
  {
        // same as crate_list
    }

struct node *insertion_at_last (struct node *start)
    {
      // same as pseudo code for insert a new
      // node at last
    }

struct node *display_list (struct node *start)
    {
      // same as display list
    }

int main ()
{
    struct node *start=NULL;
```

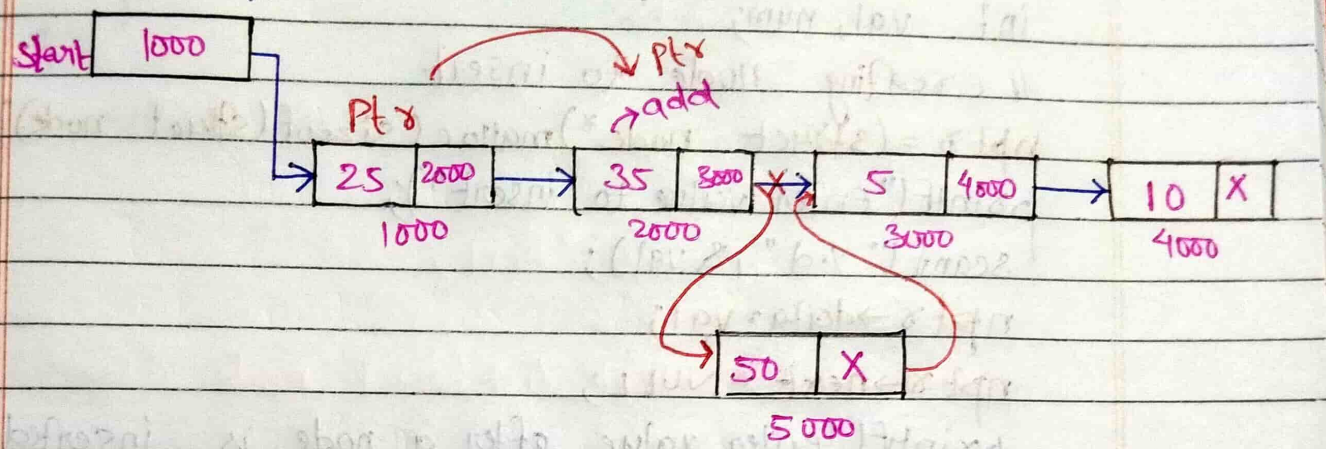
Akash

```
    start = create_list(start);
    start = insertion_at_last(start);
    start = display_list(start);
    return 0;
}
```

Akash

(iii) Insertion at a specific ~~position~~ value :

The new node is inserted after a given value. ~~position.~~



Algorithm :

Step 1. Create nptr [new_node]

2. IF (start == NULL)
   print "empty list"

3. Else.
   3.1. set ptr = start
   3.2 Repeat until ptr != NULL and ptr → data != num
   ptr = ptr → next

4. I.F ptr == NULL
   print "value not found in the list"

5. else.
   5.1. nptr → next = ptr → next
   ptr → next = nptr

6. Exit

Akash

Pseudo code for insert a new node at specific value:

```
struct node *ptr, *nptr;
int val, num;
// creating node to insert
nptr = (struct node*)malloc(sizeof(struct node));
printf("enter value to insert");
scanf("%d", &val);
nptr->data = val;
nptr->next = NULL;
printf("enter value after a node is inserted");
scanf("%d", &num);
if(start == NULL)
  {
    Printf("empty list");
  }
else
  {
    ptr=start;
    while(ptr!=NULL && ptr->data!=num)
      {
        Ptr=ptr->next;
      }
    if(ptr==NULL)
      {
        Printf("value not found in list\n");
      }
```

Akash

```
else
    {
        nptr->next = ptr->next
        ptr -> next = nptr;
    }
}
return start;
```

Akash

```
// c program for inserting a node at
// specific value

#include<stdio.h>
#include<stdlib.h>
struct node
{
    int data;
    struct node *next;
};
struct node *create_list(struct node *start)
{
    // same as create_list
}

struct node *insertion_at_specific_value(struct node *start)
{
    // same as pseudo code for insert a new node at
    // specific value
}

struct node *display_list(struct node *start)
{
    // same as display_list
}

int main()
{
    struct node *start = NULL;
```

Akash

```
    start = create_list(start);
    start = insertion_at_specific_value (start);
    start = display_list(start);
    return 0;
}
```

Akash

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